internship report

My internship at CERN

Summary

This document summarises my one-week internship in October 2023, carried out at CERN in Geneva.

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References

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List of revisions

Revision	*Date*	*Description of changes*
1.0	2023-10	

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1 Introduction

The project presented in this report was carried out during my internship week. I was fortunate to be welcomed at CERN in a section dedicated to the design of power converters, supplying the superconducting magnets of CERN's accelerators.

CERN (European Organization for Nuclear Research) is one of the largest particle physics research organisations in the world.

The main objective of CERN is to conduct fundamental research in particle physics, particularly in the field of particle acceleration and high-energy physics. CERN is famous for its particle accelerator (Figure 1), the Large Hadron Collider (LHC), which is the largest and most powerful particle accelerator in the world. CERN scientists use the LHC to study the fundamental constituents of matter, understand the forces that govern them and explore the origins of the universe.

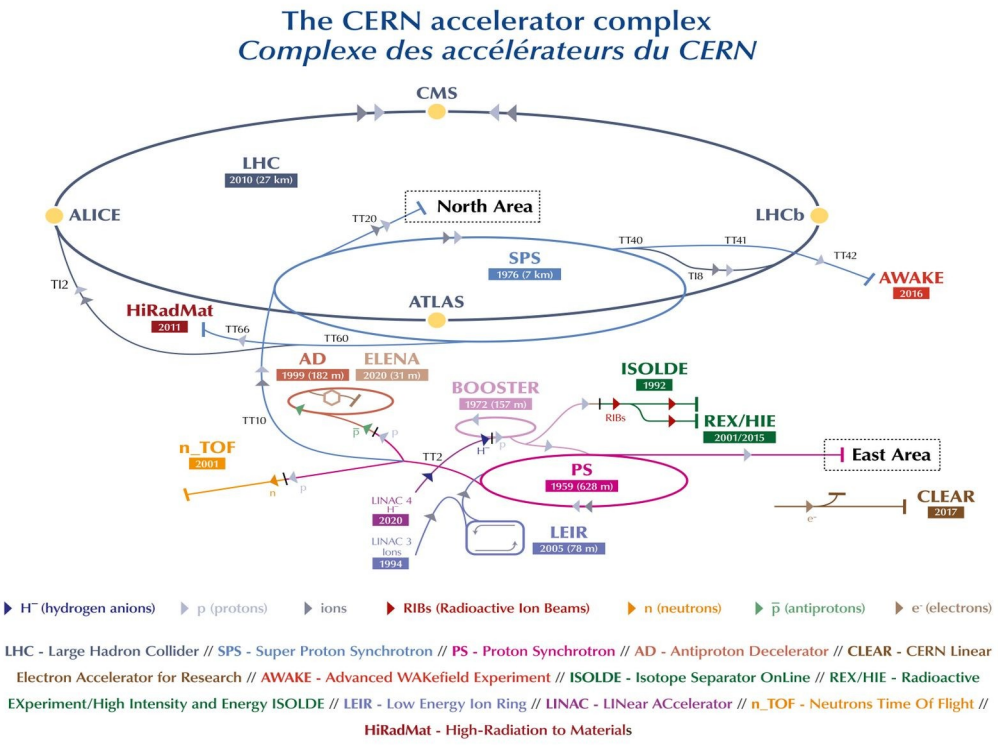


Figure 1: CERN accelerator complex

2 Internship subject

The objective of this internship week was to create a command interface to control a Transfer Function Analyser or TFA.(Figure 2)

The TFA is a widely used measurement tool because it makes it possible to know the behaviour of any system at different frequencies by measuring the gain and phase of this system over a frequency range.

The code must be able to send configuration commands to the device via the serial port, start the measurements and retrieve all the measurement results in the form of a csv file. The program is written in python and must be compatible with Linux as well as Windows.



Figure 2: GP102 TFA

3 My internship week

D	Work	Expected result
1	• Understand how the TFA works and what it is used for • Getting to grips with the TFA • Understanding how a serial port works. • Using an oscilloscope to see the sweep performed by the TFA while it is running.	Be able to estimate the time required for the next steps. Getting to grips with the TFA.
2	• Searching for the commands to configure the TFA using its documentation (> 100 pages, in English – typical commands in Figure 3) • Searching for the commands to communicate with and control the TFA	Be able to send a command and receive simple information from the TFA.
3	• Creating the graphical interface (Error: Reference source not found , Figure 5) • Implementing a few basic user-oriented functions (signal viewer)	Having a working graphical interface able to send commands to the TFA
4	• Creating all the functions to control the TFA • Decoding the data sent by the TFA and inserting it into a csv file	Having a program able to meet the initial expectations
5	• Improving the graphical interface by disabling certain features to optimise the user experience • Testing the program on Linux and Windows	The program must work on Linux as well as Windows

3.1 Program preview

```
1 AMPLIT,2.0
2 FREQUE,1000
3 OFFSET,1
4 WAVEFO,SINEWA
5 MODE,GAINPH
6 FSWEEP,50,10,1000
7 SPEED,MEDIUM
8 FILTER,NORMAL,FIXED
9 RANGE,CH1,MANUAL,3V
10 COUPLI,CH1,AC+DC
11 SCALE,CH1,10
12 RANGE,CH2,MANUAL,3V
13 COUPLI,CH2,AC+DC
```

Figure 3: commands (preview)

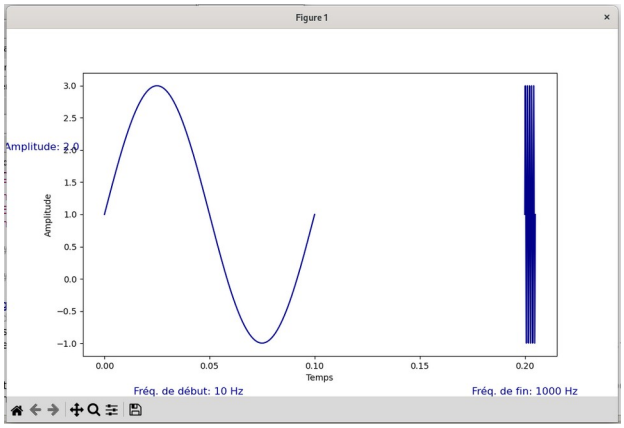


Figure 4: View sinusoidal signal button

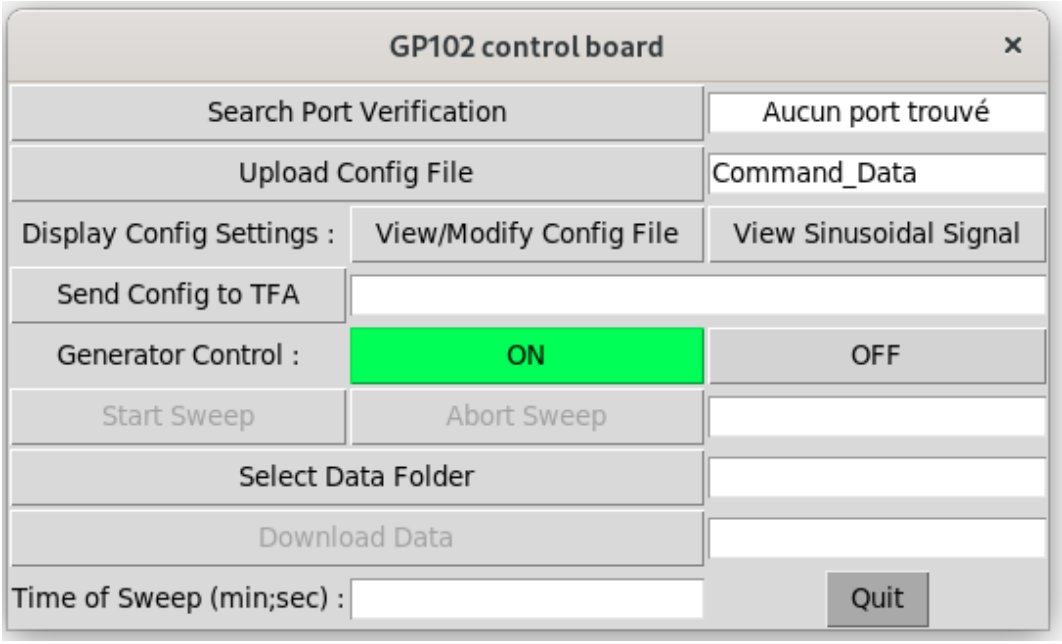


Figure 5: Graphical interface

4 Conclusion

This internship was a very enriching experience for me: it allowed me to be placed in a professional setting, to face complex and unfamiliar problems (discovering the Transfer Function Analyser device), and to realise the importance of organisation at work and time management, as some tasks turned out to be longer and more difficult than they seemed. The result lived up to expectations, with a perfectly working code delivered (500 lines), presented in the form of function blocks.